

NGUYEN PHAN BAO VIET

SENIOR SOFTWARE QA/QC ENGINEER | AUTOMATION & MANUAL SPECIALIST

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 Online Portfolio

PROFESSIONAL SUMMARY

- **Senior Software QA/QC Engineer with nearly 5 years of comprehensive experience** across the entire Software Testing Life Cycle (STLC), specializing in high-reliability **Banking & Fintech**, **Enterprise HRM**, and **AI-powered EdTech** platforms.
- **Hands-on expertise in Automation Testing:** designing, architecting, and maintaining scalable End-to-End (E2E) automation frameworks using **Playwright (TypeScript)**, Page Object Model (POM), Data-Driven Testing (DDT), and CI/CD pipelines.
- **Adept at dual-track quality engineering:** seamlessly balancing practical **Manual Testing** (exploratory, edge-case analysis, boundary value, risk-based verification) and robust **Automation Testing** to maximize test coverage and shorten release cycles.
- **Proficient in API & Database Verification:** experienced in RESTful API testing (Postman, Playwright APIRequestContext), payload validation, and formulating **complex SQL queries** (PostgreSQL, MySQL - joins, subqueries, transaction logs) and NoSQL (MongoDB aggregation pipelines).
- **High-performing Agile/Scrum collaborator with fluent English communication:** confident in conducting sprint reviews, defect triage, and technical discussions directly with developers, product owners, and international stakeholders.

CORE COMPETENCIES & TECHNICAL SKILLS

Testing & Methodologies	Manual Testing, Automation Testing (Playwright), Functional Testing, Integration Testing, Regression Testing, API Testing, Data-Driven Testing (DDT), Test Case Design, Defect Lifecycle, Agile/Scrum.
Databases & Backend	SQL (PostgreSQL, MySQL - complex queries, joins, transactions), NoSQL (MongoDB aggregation), RESTful APIs, Node.js.
CI/CD & Tools	GitHub Actions, Docker, Git & GitHub, Jira, Confluence, Postman, Playwright Test Runner.
Domain Expertise	EdTech & AI (Adaptive Learning, Knowledge Gap Diagnosis), Banking & Payment Networks (Clearing & Settlement, Credit Scoring), Enterprise HRM & Payroll .
Languages	English (IELTS 5.5 - 6.0: Can read technical documents, write emails, formal reports & participate in technical discussions), French (DELF B1 - B2 level), Vietnamese (Native).

PROFESSIONAL WORK EXPERIENCE

IT DRAGONS	Da Nang, Vietnam
Senior Software Quality Control (QC) Engineer	06/2022 – Present

1. Enterprise HRM & Payroll System

PLAYWRIGHT AUTOMATION

SENIOR QC ROLE

Role: Senior QC Engineer | Testing Approach: Playwright Automation & Manual Testing

DATA-DRIVEN TESTING (DDT)

- Led the overall testing lifecycle; architected comprehensive Test Strategies and E2E automation frameworks using **Playwright (TypeScript)** with **Data-Driven Testing (DDT)**.
- Automated payroll computation verification across complex tax brackets, social insurance deductions, overtime multipliers, and allowances.
- Automated cross-browser testing across Chromium, Firefox, and WebKit; implemented automated **RBAC security test scripts** preventing unauthorized data leaks of salary records.
- Executed data validation testing on the intermediate Hub using **complex SQL queries**, ensuring zero data corruption between raw client imports and backend databases.

2. Banking Payment Network

PLAYWRIGHT AUTOMATION

MIDDLE QC ROLE

Role: Middle QC Engineer | Testing Approach: Playwright Automation & Manual Testing

FINTECH & BANKING

- Designed and maintained a **Playwright (TypeScript)** E2E Automation Framework following the Page Object Model (POM) pattern.
- Automated core authentication: user **Login, Logout**, session persistence, and invalid credential error handling.
- Automated **new merchant onboarding** flow: registration, business validation, and merchant portal activation.
- Automated **purchase transaction workflow**: payment initiation, card/account input validation, and payment gateway callback verification.
- Automated **transaction hold features** (pre-authorized funds hold reservation, expiration, and status transitions) and **refund workflows** (full/partial refunds, ledger reversal verification).
- Wrote targeted **SQL queries** to audit transaction status flags and refund logs in the database; executed manual testing for third-party network timeout edge cases.

3. AI-Powered Adaptive Learning & Knowledge Assessment Platform (EdTech)

100% MANUAL TESTING

MIDDLE QC ROLE

Role: Middle QC Engineer | Testing Approach: 100% Manual Testing & API/Data Validation

API & DATA VALIDATION

- Formulated master **Test Strategy and Test Plans** covering syllabus/past exam ingestion, dynamic examination UI, error-pattern detection engine, and teacher diagnostic dashboards.
- Performed manual functional, exploratory, and boundary testing on real-time exam interfaces (autosave, timer sync, submission integrity).
- Conducted extensive **API testing using Postman** for AI microservices (document parsing, prompt-to-question generation, token thresholds, JSON schema validation).
- Validated AI Error-Pattern Recognition & Knowledge Gap Engine**: designed simulation matrices for recurring arithmetic slips vs. thematic theory errors, verifying accurate gap classification and remediation plans.
- Executed **complex SQL & MongoDB queries** to audit student telemetry event logs and knowledge gap matrices.

4. Credit Scoring & Financial Risk Assessment System

100% MANUAL TESTING

MIDDLE QC ROLE

Role: Middle QC Engineer | Testing Approach: 100% Manual Testing & Data Validation

BANKING DOMAIN

- Planned and executed comprehensive manual testing strategies for multi-variable credit scoring algorithms and rule engines.
- Conducted rigorous **Data Validation and Negative Testing** with corrupted or anomalous credit bureau records to evaluate system error-handling resilience.
- Formulated **complex SQL queries** to audit score computation tables, validation rules, and compliance audit logs.

- Analyzed business requirements, designed detailed test cases and test scenarios for an internal CRM web application.
- Executed manual functional, UI/UX, and regression testing across sprint cycles to identify defects and ensure requirement alignment.
- Performed basic API testing using Postman to validate CRUD endpoints, HTTP status codes, and JSON response payloads.
- Wrote basic SQL queries (SELECT, WHERE, JOIN) to prepare test data and verify database integrity.
- Logged, tracked, and verified defect lifecycles in Jira; collaborated with development team in Agile standups.
- **Skills utilized:** Functional Testing, Regression Testing, Test Case Design, API Testing (Postman), SQL, Jira, Agile/Scrum.

EDUCATION

Web Fullstack Developer Certificate Program

02/2020 – 02/2022

FUNiX

- Completed the FUNiX Web Fullstack Developer Certificate Program.
- GPA: 8.4 / 10